FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 06

By - Dr. Puneet Sharma Sir



Recap of previous lecture

" MENSURATION "



Topics to be covered

"CONIC SECTION"



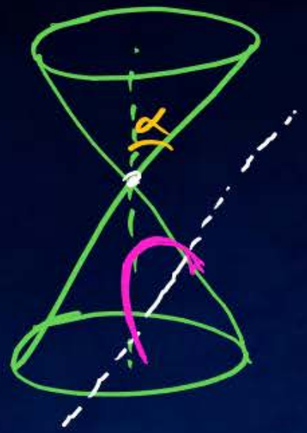
(1)



if $0 < \theta < \alpha$
& Plane passes through vertex then we get Pair of st. lines



if $\theta = \frac{\pi}{2}$
Circle



$\theta = \alpha$
Parabola



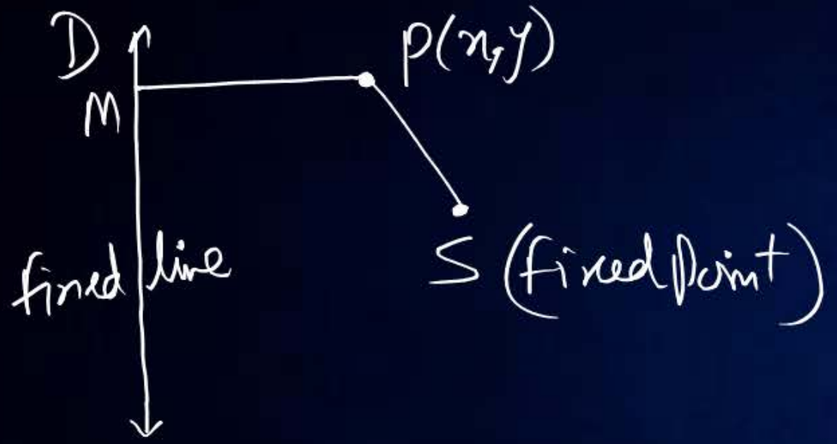
$0 < \theta < \alpha$
Hyperbola



$\alpha < \theta < \frac{\pi}{2}$
Ellipse

Plane does not pass through vertex

(2) Analytic Defⁿ –



Conic section is the Locus of point $P(x,y)$ which moves in such a manner that Ratio of it's distance from fixed Point 'S' and fixed line DD' is always constant i.e

$$\frac{PS}{PM} = \text{Constant}(e)$$

① for Parabola, $e=1$
② for Ellipse, $e<1$
③ for Hyperbola, $e>1$
④ for circle, $e=0$

Note Consider two straight lines $a_1x+b_1y+c_1=0$ & $a_2x+b_2y+c_2=0$

then Combined eqⁿ $(a_1x+b_1y+c_1)(a_2x+b_2y+c_2)=0$ is called Eqⁿ of pair of St. lines.

(3) General equⁿ of two Degree → it is given as; $\boxed{ax^2+by^2+2hxy+2gx+2fy+c=0}$

Here we will represent the Value of $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ by Δ i.e $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ ①

① Equⁿ ① will represent, Pair of straight lines if $\Delta = 0$

② " " " Circle if $\Delta \neq 0$, $a=b$, $h=0$ (i.e coeff of $xy=0$)

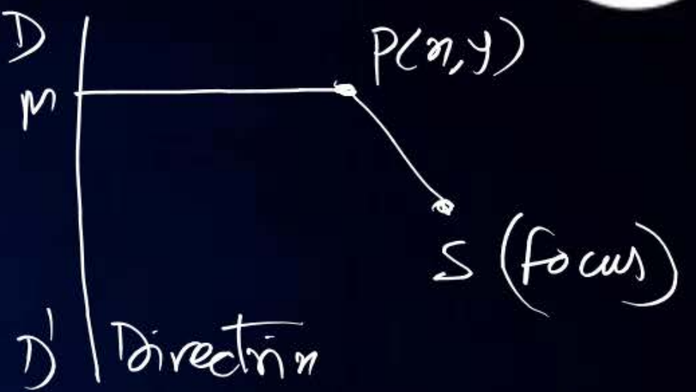
③ " " " Parabola if $\Delta \neq 0$ & $h^2=ab$

④ " " " Ellipse if $\Delta \neq 0$ & $h^2<ab$

⑤ " " " Hyperbola if $\Delta \neq 0$ & $h^2>ab$

⑥ " " " Rectangular Hyperbola if $\Delta \neq 0$, $h^2>ab$ & $a+b=0$

Defⁿ of Some Basic terms → $\boxed{\frac{PS}{PM} = e}$ where S is Fixed Point & DD' = Fixed line



(1) Focus → Fixed Point is called focus 'S'

(2) Directrix → Fixed line is called Directrix 'DD'

(3) Constant (e) → that Constant Ratio is called Eccentricity

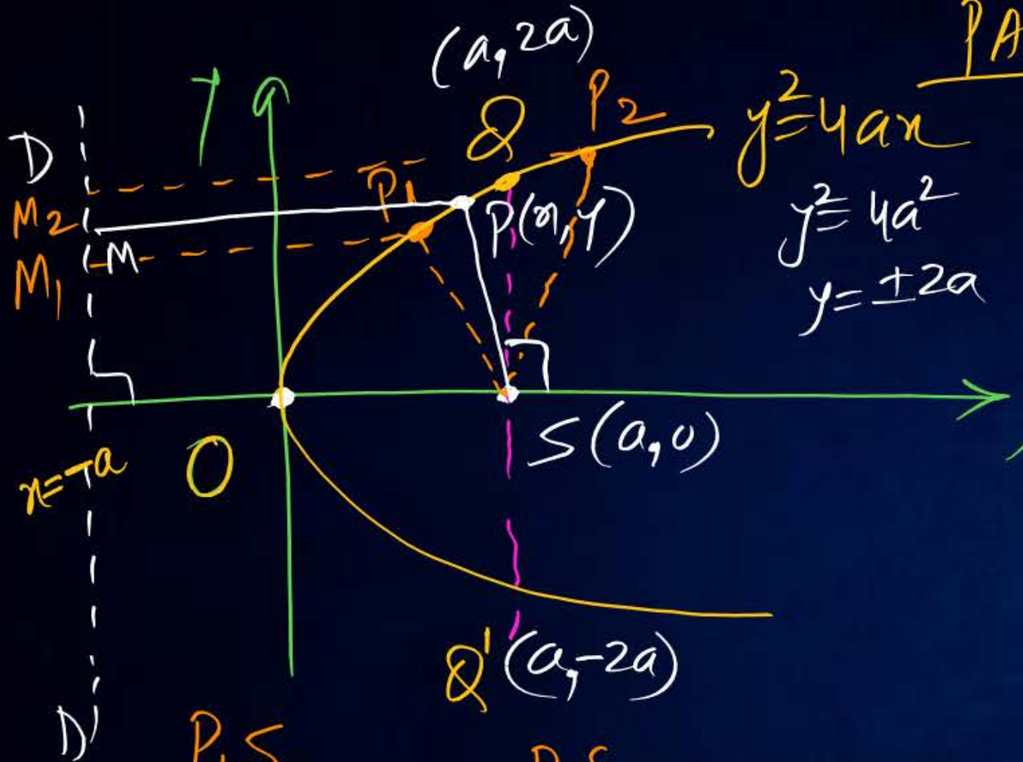
(4) Axis → Line passes through the focus (S) and ⊥ʳ to the Directrix (DD') is called Axis

(5) Vertex → the point of intersection of Conic with Axis is called Vertex.

(6) Latus Rectum → it is the Chord of Parabola, passes through the focus & ⊥ʳ to it's Axis

(7) Focal length → the Distance of any point P(x,y) from focus is called Focal length.

PARABOLA



Focus : $(a, 0)$

Directrix : $x + a = 0$

Axis : X axis ie $y = 0$

Vertex : $(0, 0)$

Eqn of LR (QQ') : $x - a = 0$

Length of LR : $4a$

$ax + by + c$ & $P(x_1, y_1)$

Length of $\perp^r$ $= \dfrac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}$

$\dfrac{P_1S}{P_1M_1} = 1, \dfrac{P_2S}{P_2M_2} = 1, \ldots\ldots$

ie $P_1S = P_1M_1, P_2S = P_2M_2, \ldots\ldots$

Focal length of $P(x, y)$ →

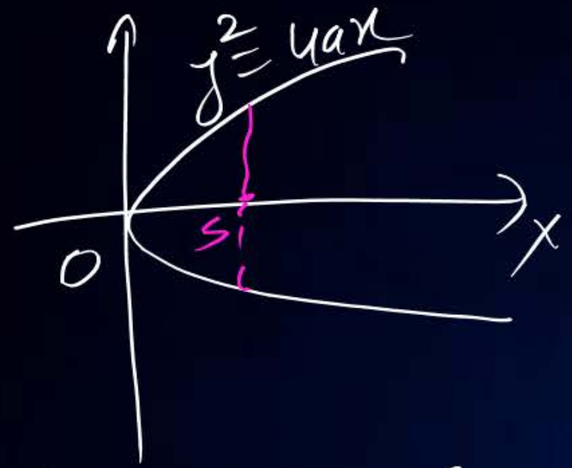
$PS^2 = (x-a)^2 + (y-0)^2 = x^2 + a^2 - 2ax + y^2$

$= x^2 + a^2 + 2ax = (a+x)^2 \Rightarrow PS = (a+x)$

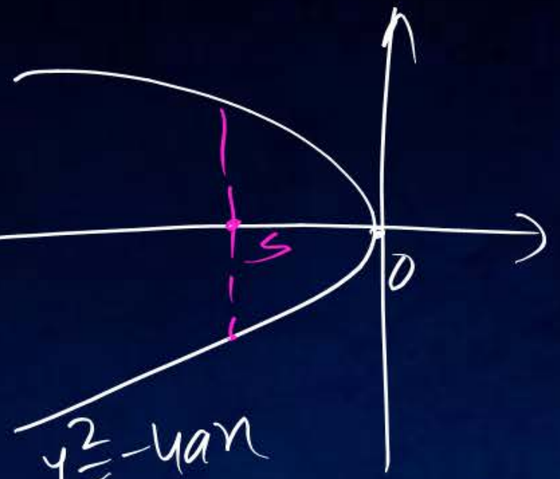
Proof of $y^2 = 4ax$ → $PS = PM \Rightarrow PS^2 = PM^2$

$(x-a)^2 + (y-0)^2 = \left(\dfrac{x+a}{\sqrt{1^2+0}}\right)^2$

$x^2 + a^2 - 2ax + y^2 = x^2 + a^2 + 2ax \Rightarrow y^2 = 4ax$

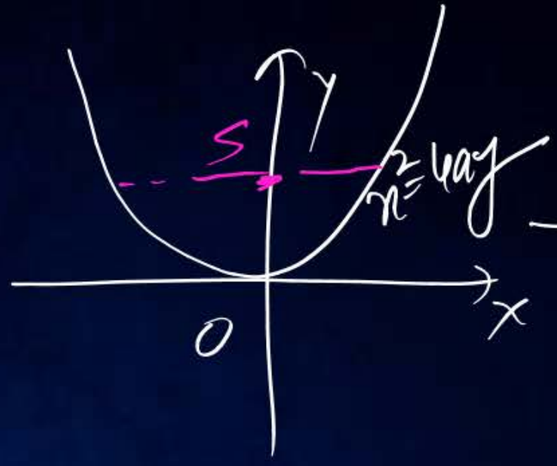


Opening Rightward

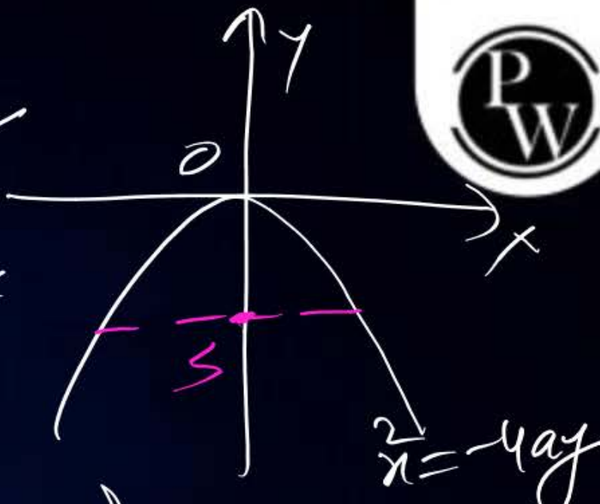


$y^2 = -4ax$

Opening Leftward



Opening upward



Opening Downward

	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Focus	$(a,0)$	$(-a,0)$	$(0,a)$	$(0,-a)$
eqn of Directrix	$x=-a$	$x=a$	$y=-a$	$y=a$
Axis	X axis	X axis	y axis	y axis
Vertex	$(0,0)$	$(0,0)$	$(0,0)$	$(0,0)$
eqn of LR	$x=a$	$x=-a$	$y=a$	$y=-a$
length of LR	$4a$	$4a$	$4a$	$4a$
F. length	$a+x$	$a-x$	$a+y$	$a-y$

eg: Convert $y^2 - 8y - x + 19 = 0$ into standard form & find various terms

Sol:- $y^2 - 8y = x - 19$

$y^2 - 2(4)y + 16 - 16 = x - 19$

$(y-4)^2 = x - 19 + 16$

$(y-4)^2 = (x-3)$ —— (1)

Let $x-3 = X$ & $y-4 = Y$ then $\boxed{Y^2 = 1X}$ —— (2)

(1) Focus: $Y^2 = 4aX \Rightarrow 4a = 1 \Rightarrow a = \frac{1}{4}$

So Focus is $(a,0)$ i.e $\left.\begin{matrix} X=a \\ Y=0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} x-3=\frac{1}{4} \\ y-4=0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} x=\frac{13}{4} \\ y=4 \end{matrix}\right.$ i.e Focus is $\left(\frac{13}{4}, 4\right)$

Axis:- is X axis i.e $y=0 \Rightarrow \boxed{y=4}$

Vertex:- $(0,0)$ $\left.\begin{matrix} X=0 \\ Y=0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} x-3=0 \\ y-4=0 \end{matrix}\right\} \Rightarrow \left.\begin{matrix} x=3 \\ y=4 \end{matrix}\right.$

i.e Vertex $= (3,4)$

Directrix: $X = -a \Rightarrow x-3 = -\frac{1}{4} \Rightarrow x = 3 - \frac{1}{4} = \frac{11}{4}$

i.e $x = \frac{11}{4}$ is the equ^n of Directrix

equ^n of LR:- $X = a \Rightarrow x-3 = \frac{1}{4} \Rightarrow x = \frac{13}{4}$

Length of LR:- $= 4a = 4\left(\frac{1}{4}\right) = 1$ unit

Verification :- $1.\ y^2 = 8y - x + 19 = 0$

on Comparision with

$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$

$\Rightarrow a = 0,\ b = 1,\ h = 0,\ g = -\frac{1}{2},\ f = -4,\ c = 19$

$\because \boxed{h^2 = ab}$ i.e given eqⁿ is parabola

$\& \Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1/2 \\ 0 & 1 & -4 \\ -\frac{1}{2} & -4 & 19 \end{vmatrix}$

$= 0 - 0 + \left(-\frac{1}{2}\right)\left[0 + \frac{1}{2}\right] = -\frac{1}{4}$

$\Delta \neq 0$ ☺

eg $\boxed{x^2 = 6y}$: $x^2 = 4ay$

$4a = 6 \Rightarrow a = \frac{6}{4} = \frac{3}{2}$

Focus $= (0, a) = \left(0, \frac{3}{2}\right)$, Vertex $= (0, 0)$

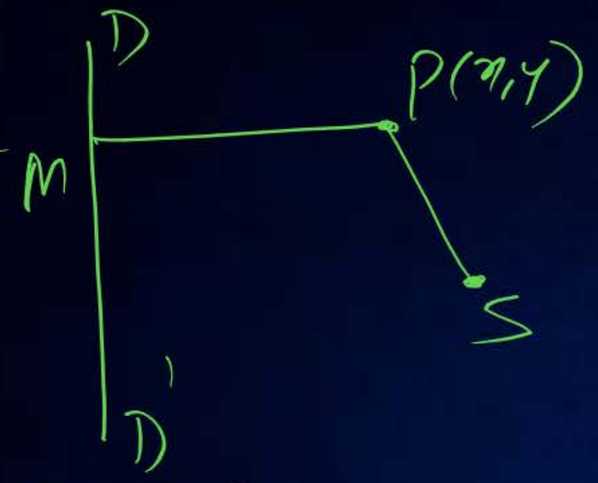
eqⁿ of Dirᵗ : $y = -3/2$

$\because$ LR : $y = 3/2$

Length of LR : $4a = 6$ Ans

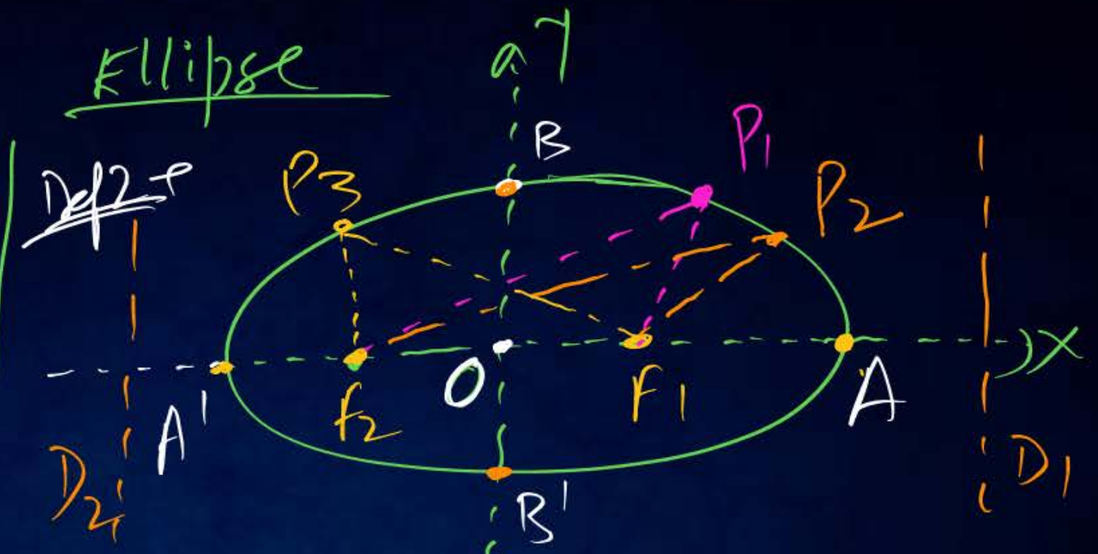
Ellipse

Def 1:



$$\frac{PS}{PM} = e \quad (\text{where } e < 1)$$

Def 2:

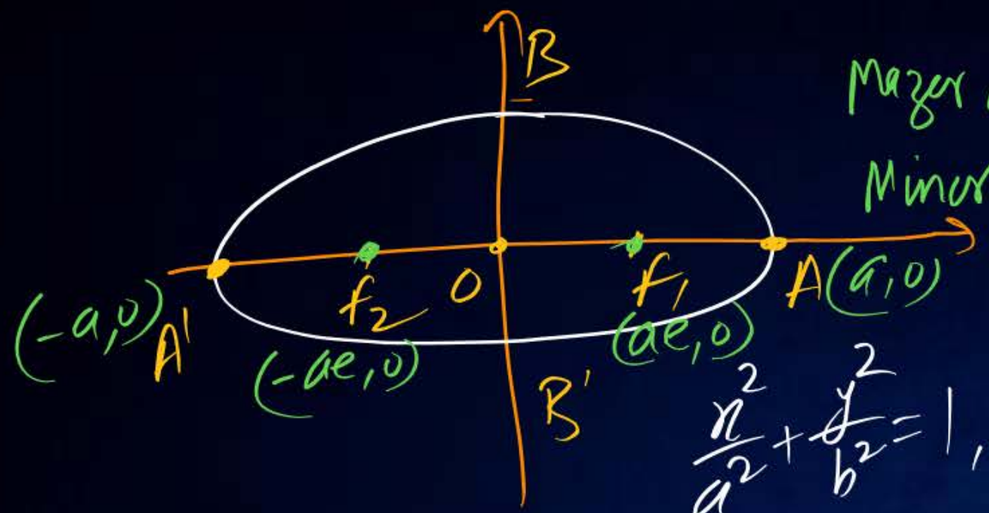


$$P_1f_1 + P_1f_2 = P_2f_1 + P_2f_2 = P_3f_1 + P_3f_2 = AA'$$

i.e sum of the distances of any Random Point 'P' from two Fixed Points f_1 & f_2 is always constant i.e $\boxed{Pf_1 + Pf_2 = \text{Length of Mayor Axis}}$

where f_1 & f_2 are Foci, Mayor Axis $= AA'$, Centre $= (0,0)$

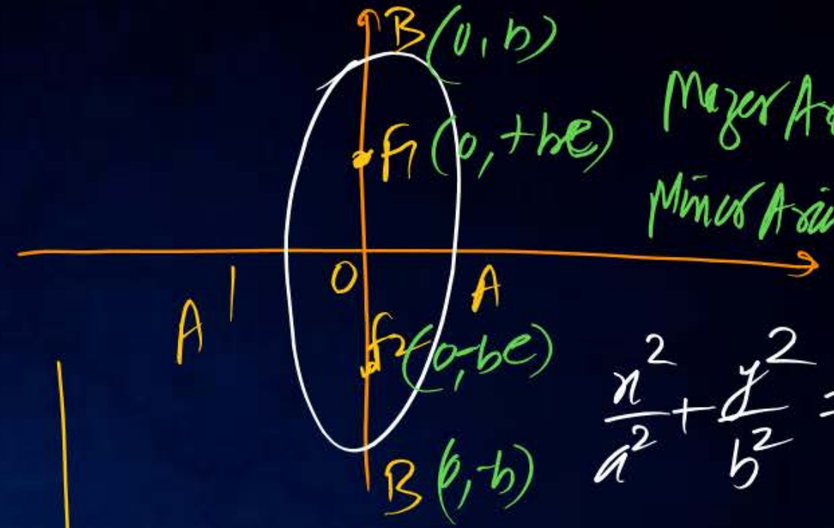
A & A' are Vertices, Minor Axis $= BB'$,



Major Axis = AA'

Minor " = BB'

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $(a>b)$



Major Axis = BB'

Minor Axis = AA'

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $(a<b)$

Centre	$(0,0)$	$(0,0)$
Vertices	$(\pm a,0)$	$(0,\pm b)$
foci	$(\pm ae,0)$	$(0,\pm be)$
Eccentricity	$b^2=a^2(1-e^2)$	$a^2=b^2(1-e^2)$
eqn of Directrix	$x=\pm\frac{a}{e}$	$y=\pm\frac{b}{e}$
Length of LR	$2b^2/a$	$2a^2/b$

	(for $a>b$)	(for $a<b$)
Length of Major axis (AA')	$2a$	$2b$
eqn of " "	x axis i.e $y=0$	y axis i.e $x=0$
Length of Minor axis	$2b$	$2a$
eqn of Minor axis (BB')	y axis i.e $x=0$	x axis i.e $y=0$

eg: Find the equ^n of ellipse for which vertices are $(\pm 5, 0)$ & foci $(\pm 4, 0)$

Sol: Vertices: $(\pm a, 0) \simeq (\pm 5, 0) \Rightarrow a = 5$

Foci: $(\pm ae, 0) \simeq (\pm 4, 0) \Rightarrow ae = 4$

ie $e = \frac{4}{a} = \frac{4}{5}$

Now $b^2 = a^2(1-e^2)$

$b^2 = 25\left(1 - \frac{16}{25}\right) = 9 \Rightarrow b = 3$

& equ^n of Ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow \frac{x^2}{25} + \frac{y^2}{9} = 1$

eg: Convert $16x^2 + 25y^2 = 400$ into S-Form & find various Coordinates & e=?

Sol: $\frac{16x^2}{400} + \frac{25y^2}{400} = 1 \Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$ ——— (1)

$\Rightarrow a^2 = 25 \Rightarrow a = 5$, $b^2 = 16 \Rightarrow b = 4$

W.k.t that; $b^2 = a^2(1-e^2) \Rightarrow 16 = 25(1-e^2)$

$1 - e^2 = \frac{16}{25} \Rightarrow e^2 = 1 - \frac{16}{25} = \frac{9}{25} \Rightarrow e = \frac{3}{5}$

Length of LR $= 2b^2/a = \frac{2 \times 16}{5} = 32/5$

Vertices $= (\pm a, 0) = (\pm 5, 0)$

foci $= (\pm ae, 0) = (\pm 3, 0)$

Directrix: $x = \pm \frac{a}{e} = \pm \frac{5}{3/5} \Rightarrow x = \frac{25}{3}$

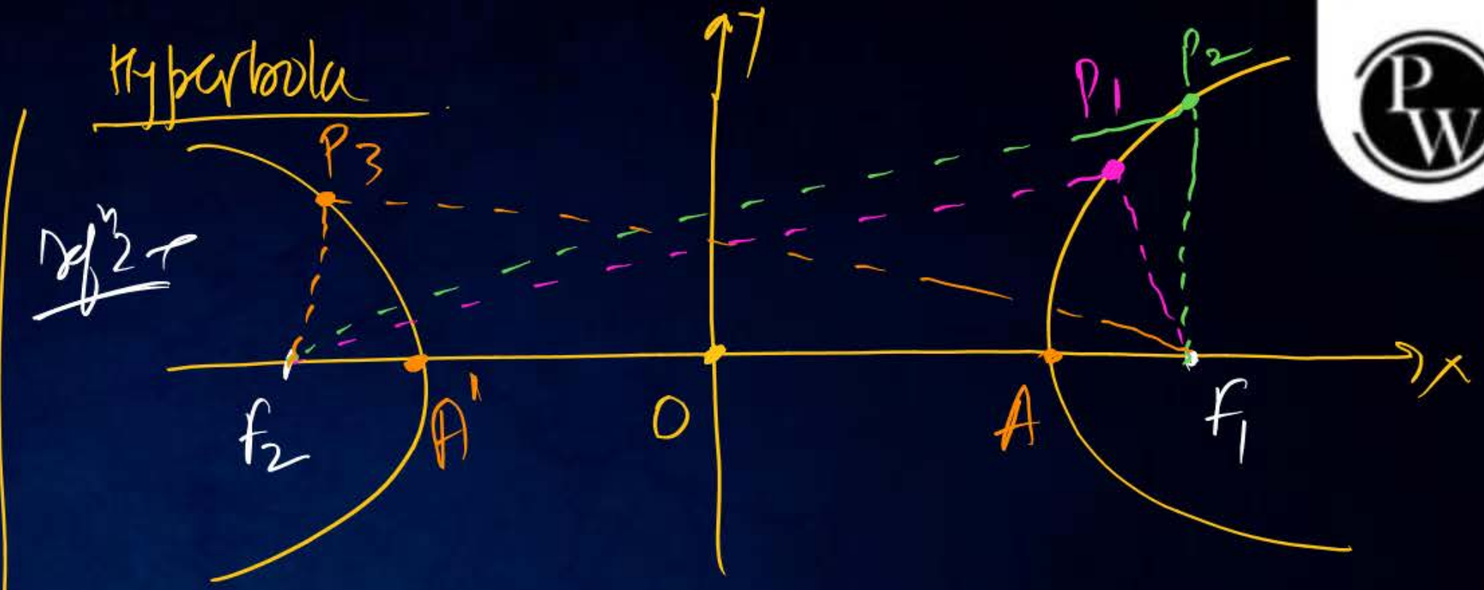
Hyperbola

Def 1 :-



$$\frac{PS}{PM} = e \quad : (e > 1)$$

Def 2 :-



$$P_1f_1 \sim P_1f_2 = P_2f_1 \sim P_2f_2 = P_3f_1 \sim P_3f_2 = AA'$$

i.e Difference of the distances of any Point P(x,y) from two fixed Points f_1 & f_2 is always Constant i.e

$$Pf_1 \sim Pf_2 = \text{Length of Transverse Axis } (AA')$$

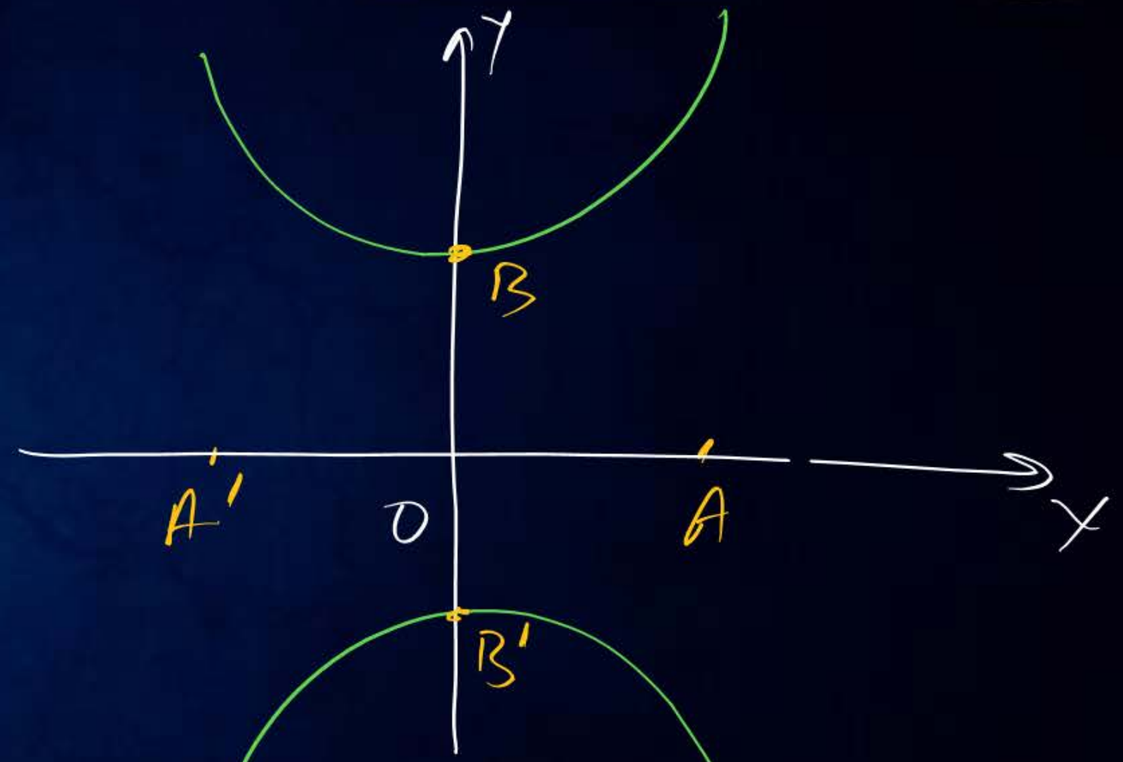
Hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Conjugate Hyperbola: $\frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$



Transverse Axis = AA' & length = $2a$

Conjugate Axis = BB' & length = $2b$



T-Axis = BB' & length = $2b$

C-Axis = AA' & length = $2a$

	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
Centre	$(0,0)$	$(0,0)$
Vertices	$(\pm a, 0)$	$(0, \pm b)$
Foci	$(\pm ae, 0)$	$(0, \pm be)$
eqn of Directrix	$x = \pm \frac{a}{e}$	$y = \pm b/e$
Length of L-R	$2b^2/a$	$2a^2/b$
eqn of T axis (AA')	X axis i.e $y=0$	Y axis i.e $x=0$
Length " " "	$2a$	$2b$
eqn of C. Axis (BB')	Y axis is $x=0$	X axis i.e $y=0$
Length of C "	$2b$	$2a$
Eccentricity	$b^2 = a^2(e^2-1)$	$a^2 = b^2(e^2-1)$

Q: find the equ^n of Hyperbola for which vertices $(\pm 5, 0)$ & foci $(\pm 7, 0)$

Sol: Vertices $= (\pm a, 0) = (\pm 5, 0) \Rightarrow \boxed{a = 5}$

Foci $= (\pm ae, 0) = (\pm 7, 0) \Rightarrow ae = 7$

ie $e = \frac{7}{a} = \boxed{\frac{7}{5}}$

Now $b^2 = a^2(e^2 - 1) = 25\left(\frac{49}{25} - 1\right) = 24$

Hence equ^n of Hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\frac{x^2}{25} - \frac{y^2}{24} = 1$$

Q: convert $16x^2 - 9y^2 = 144$ into Standard form and evaluate e as well as Length of L-R.

Sol: $\frac{16x^2}{144} - \frac{9y^2}{144} = 1 \Rightarrow \boxed{\frac{x^2}{9} - \frac{y^2}{16} = 1}$ — (1)

$a = 3, b = 4$

Now $b^2 = a^2(e^2 - 1) \Rightarrow e^2 = \frac{b^2}{a^2} + 1 = \frac{16}{9} + 1 = \frac{25}{9} \Rightarrow \boxed{e = \frac{5}{3}}$

Length of L-R $= 2b^2/a = \frac{2(16)}{3} = 32/3$

Vertices $= (\pm a, 0) = (\pm 3, 0)$

foci $= (\pm ae, 0) = (\pm 5, 0)$

Directrix: $x = \pm \frac{a}{e} = \pm \frac{3}{5/3} = \pm \frac{9}{5}$

Length of T.Axis $= 2a = 6$

& C.Axis $= 2b = 8$

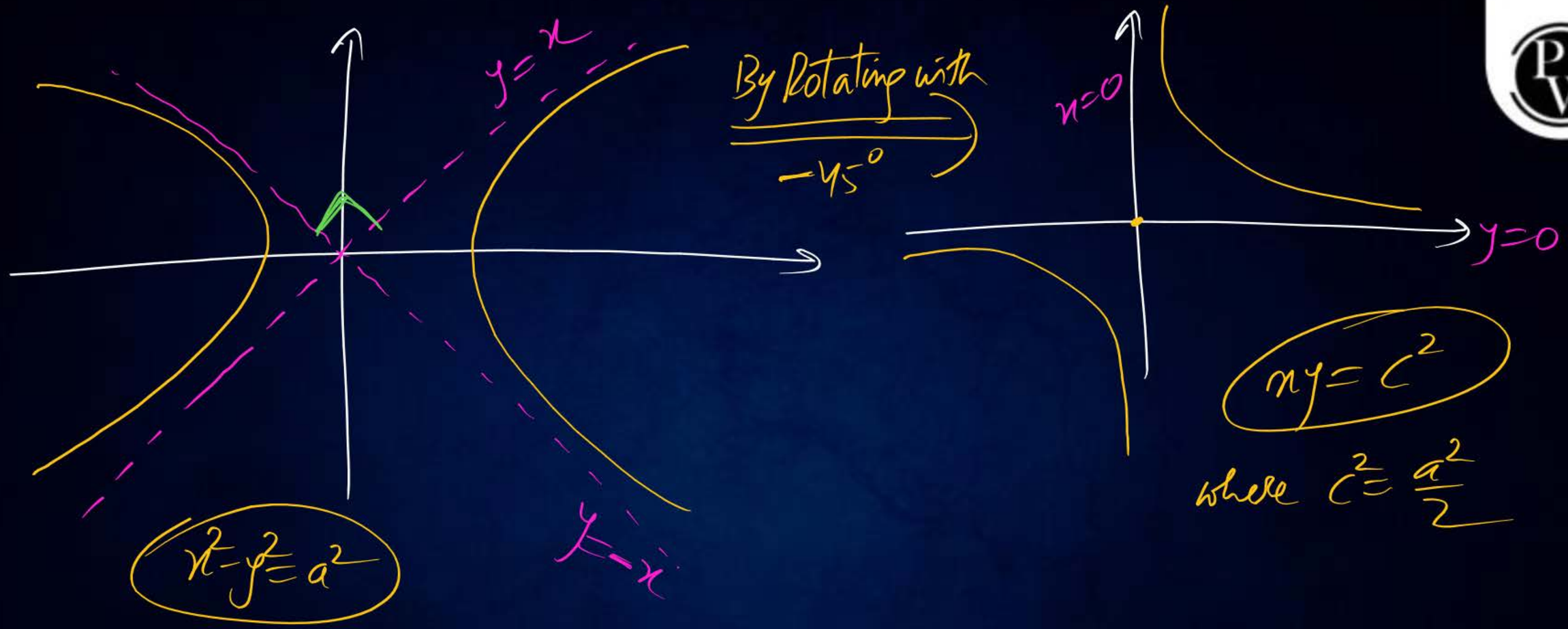
Rectangular Hyperbola:- If asymptotes of Hyperbola are $\perp^{r}$ to each other then such type of Hyperbola is called Rectangular Hyperbola:

and for this we can take $a=b$ in the standard equ^{n} i.e. In $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

we get $\frac{x^2}{a^2}-\frac{y^2}{a^2}=1$ or $\boxed{x^2-y^2=a^2}$ S. form of R. Hyperbola

Asymptotes of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ are $y=\pm\frac{b}{a}x$

So " of $x^2-y^2=a^2$ are $y=\pm x$ $\Rightarrow m_1=1$, $m_2=-1$ $\because m_1m_2=-1$ i.e. $y=x$ & $y=-x$ are at 90°


y = x
y = −x
x² − y² = a²
By Rotating with
−45°
x = 0
y = 0
xy = c²
where c² = a²/2

CIRCLE

(1)



$CP = r$

$CP^2 = r^2$

$$(x-h)^2 + (y-k)^2 = r^2$$

Standard form of Circle with → Centre = (h, k), rad = r

eg Centre $(3, -2)$ & rad = 5 then equ^n of circle is ?

Sol:

$$(x-3)^2 + (y-(-2))^2 = 5^2$$

$$(x-3)^2 + (y+2)^2 = 25$$

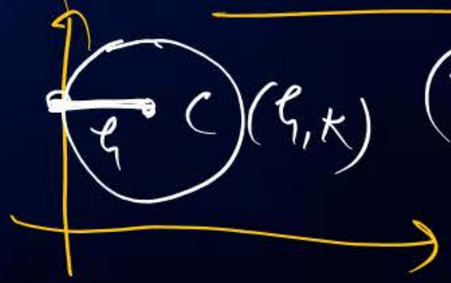
eg Centre $(-5, 4)$, rad = 2 then equ^n of circle is $(x+5)^2 + (y-4)^2 = 2^2$ //

Case I: Circle touches X axis →



$$(x-h)^2 + (y-k)^2 = k^2$$

Case II → Circle touches Y axis →



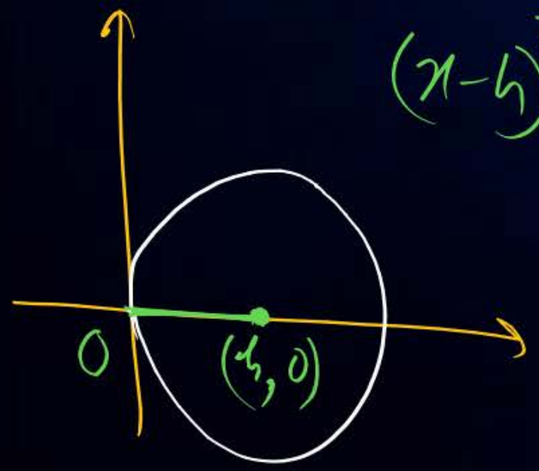
$$(x-h)^2 + (y-k)^2 = h^2$$

Case III: Circle touches both axes →



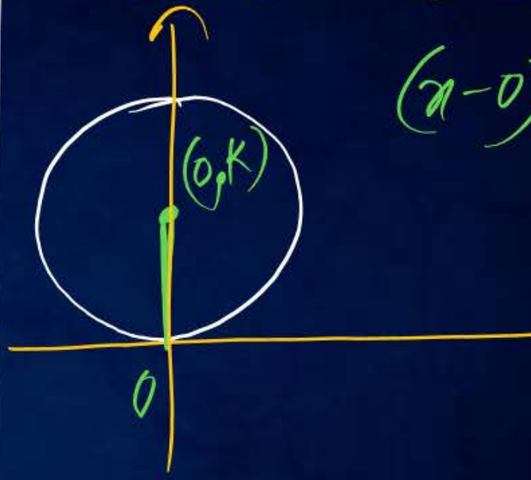
$$(x-a)^2+(y-a)^2=a^2$$

Case IV: Centre lies on x axis & passing through origin →



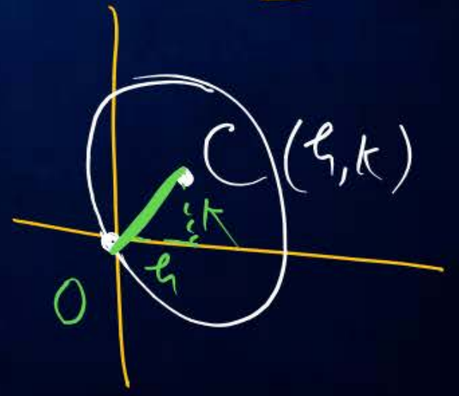
$$(x-h)^2+(y-0)^2=h^2$$

Case V: Centre lies on y axis & passing through origin →



$$(x-0)^2+(y-k)^2=k^2$$

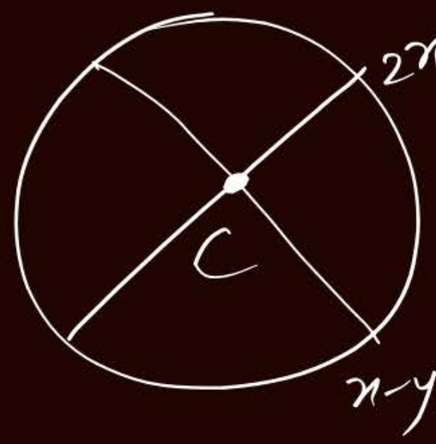
Case VI: Circle passing through origin →



$$OC=\sqrt{h^2+k^2}=\text{rad.}$$

$$(x-h)^2+(y-k)^2=(h^2+k^2)$$

eg: If two diameters of circle are given as $x-y=5$ & $2x+y=4$ & radius is of 5 units then find it's equ^{n}s



Intersecting Point $= (3,-2)$

$2x+y=4$

$x-y=5$

$3x=9$

$x=3$

$y=-2$

equ^n of circle is

$(x-3)^2+(y+2)^2=5^2$

Sp. Case → General equ^n of Circle ←

$$x^2+y^2+2gx+2fy+c=0 \quad \text{---(1)}$$

Here Centre $=(-g,-f)$ & rad $=\sqrt{g^2+f^2-c}$

eg: $x^2+y^2-6x+4y-12=0$ ---(1)

$ax^2+by^2+2hxy+2gx+2fy+c=0$

$\Rightarrow a=1, b=1, h=0, g=-3, f=+2, c=-12$

$\because a=b$ & $h=0$ so (1) is the circle

Centre $=(-g,-f)=(3,-2)$

rad $=\sqrt{g^2+f^2-c}=\sqrt{9+4+12}=5$

(M-II) $x^2+y^2-6x+4y-12=0$

$x^2-2(3)x + y^2+2(2)y-12=0$

$x^2-2(3)x+9-9 + y^2+2(2)y+4-4-12=0$

$(x-3)^2+(y+2)^2-25=0$

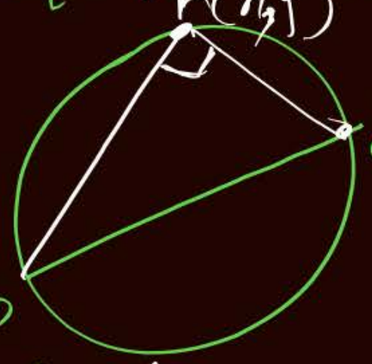
$(x-3)^2+(y+2)^2=(5)^2$

$\Rightarrow (x-h)^2+(y-k)^2=r^2$

$\Rightarrow$ Centre $=(h,k)=(3,-2)$ & rad $=5$

Another Sp. Case –

eqn of Circle in Diameter Form –



(x_1,y_1) (Slope of AP).(Slope of AQ) $=-1$

$\left(\frac{y-y_1}{x-x_1}\right)\cdot\left(\frac{y-y_2}{x-x_2}\right)=-1$

$(y-y_1)(y-y_2)=-(x-x_1)(x-x_2)$

or $\boxed{(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0}$

Note: ① $(x-h)^2+(y-k)^2=r^2$ ⟨ Centre $=(h,k)$, rad $=r$

② $x^2+y^2+2gx+2fy+c=0$ ⟨ Centre $=(-g,-f)$, rad $=\sqrt{g^2+f^2-c}$

③ if Coordinates of Diameter are (x_1,y_1) & (x_2,y_2) then eqⁿ of circle is $\boxed{(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0}$

CONIC SECTION

Parametric Coordinates

① circle $x^2+y^2=r^2$ — $(r\cos\theta, r\sin\theta)$

② Parabola $y^2=4ax$ — $(at^2, 2at)$

③ Ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ — $(a\cos\theta, b\sin\theta)$

④ Hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ — $(a\sec\theta, b\tan\theta)$

learn.

⑤ $\sin^2\theta+\cos^2\theta=1$, $1+\tan^2\theta=\sec^2\theta$


Thank
YOU